Highlights

A Multi-Layer Framework for Filtering, Merit Classification, Sentence Segmentation, and Named-Entity Extraction in Brazilian Criminal Labor-Law Decisions

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- Multi-layer filtering for noisy, heterogeneous legal corpora.
- Task-specific normalization improves classification and NER compatibility.
- Conviction-focused segmentation and legal-domain NER for structured extraction.
- Reproducible benchmark design for Portuguese criminal labor-law decisions.

A Multi-Layer Framework for Filtering, Merit Classification, Sentence Segmentation, and Named-Entity Extraction in Brazilian Criminal Labor-Law Decisions

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ARTICLE INFO

Keywords:
Legal NLP
Information extraction
Named entity recognition
Judicial decisions
Modern slavery
Portuguese language resources

ABSTRACT

This paper presents a multi-layer legal NLP framework for extracting structured information from Brazilian criminal labor-law judicial decisions related to modern slavery. Starting from a heterogeneous corpus (2,581 documents), the framework applies: (i) document-type filtering to retain only final decisions, (ii) legal-text normalization with task-specific outputs for classification and NER, (iii) penal-merit classification using Legal-BERT embeddings with classical classifiers, (iv) LLM-assisted sentence segmentation for conviction-focused documents, and (v) legal-domain NER for entity extraction. We describe the full pipeline, evaluation protocol, and error analysis, and discuss practical deployment constraints, reproducibility, and ethical/legal considerations. Results indicate that layered filtering and task-oriented preprocessing substantially improve downstream extraction quality in low-resource legal settings.

1. Introduction

1.1. Problem context and motivation

- Explain the supply-chain transparency challenge in modern slavery investigations.
- Justify why judicial decisions are high-value evidence sources.

1.2. Research gap

- Existing legal NLP pipelines are often single-stage and not robust to document heterogeneity.
- Limited resources for Portuguese criminal labor-law datasets and benchmarks.

1.3. Research questions

- RQ1: How much does multi-layer filtering improve downstream extraction reliability?
- RQ2: Which merit-classification model is most robust under class imbalance?
- RQ3: How do segmentation quality and NER quality interact in the final extraction output?

1.4. Main contributions

1. End-to-end framework from raw legal documents to structured entities.
2. Task-adaptive preprocessing design with separate outputs for classification and NER.
3. Comparative evaluation of embedding-based classifiers for penal-merit prediction.
4. Practical protocol for LLM-assisted segmentation plus legal-domain NER.

2. Related Work

The proposed pipeline spans four distinct technical areas: legal document classification and filtering, sentence segmentation and structural analysis of legal texts, named entity recognition in legal and low-resource language settings, and end-to-end legal NLP frameworks. This section surveys the most relevant prior work in each area, identifying the key contributions and limitations that motivate the design choices made in the present study.

2.1. Legal document filtering and classification

The ability to distinguish legally relevant documents from heterogeneous corpora is a prerequisite for any downstream extraction task. Ariai et al. [02_02] provide a comprehensive survey of NLP tasks in the legal domain, systematically characterizing the unique challenges that complicate classification: extensive document lengths, highly specialized vocabulary, jurisdiction-dependent terminology, and the scarcity of openly available annotated datasets. The authors identify Legal Text Classification (LTC) as a well-studied but unevenly distributed subfield, with a strong concentration of work on Chinese and English legal systems

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and comparatively little attention to other languages and jurisdictions. Their identification of domain-specific fine-tuning as a critical success factor directly informs the use of Portuguese Legal-BERT and BERTimbau encoders in Stage2 of the present work.

Resck et al. [02_01] demonstrate the effectiveness of combining a transformer-based classifier with an interpretability mechanism for the identification of implicit citation relationships in Brazilian Supreme Court decisions. Although their task differs from multi-class merit classification, their architecture illustrates the viability of applying machine learning classification to large-scale Brazilian legal corpora — precisely the linguistic and institutional context addressed in the present study. The visual analytics layer they develop also highlights the practical importance of explainability in legal AI systems. Nuranti et al. [02_03] contribute a systematic, reproducible pipeline for corpus construction and NER annotation covering Indonesian judicial decisions across three hierarchical court levels. Their use of BIO tagging with more than 50 entity types, combined with Fleiss' Kappa-based inter-annotator agreement evaluation, establishes methodological best practices that inform the annotation and quality-control protocol adopted in the present work.

2.2. Legal sentence segmentation and structural analysis

Accurate segmentation of legal documents into semantically meaningful units is a precondition for reliable information extraction. Bhattacharya et al. [03_05] present DeepRhole, a deep learning system for rhetorical role labeling of sentences in Indian and UK Supreme Court case documents, assigning functional labels — such as Fact, Argument, Precedent, and Final Judgement — to individual sentences. Their experimental evaluation demonstrates that Hierarchical BiLSTM architectures substantially outperform prior CRF-based methods that relied on handcrafted features, and that domain-specific pretraining using Law2Vec consistently improves performance. Crucially, they also report that some rhetorical label distinctions (e.g., Ratio of the Decision vs. Precedent) are inherently subjective, a finding that has direct methodological implications for any segmentation protocol that must reconcile automated predictions with human expert judgment.

Sheik et al. [03_07] address sentence boundary detection (SBD) in English legal texts, demonstrating that a hybrid CNN+CRF architecture outperforms both standalone deep learning models and open-source SBD libraries, achieving more than 4% F1 improvement. Their out-of-domain evaluation on Indian legal text confirms the generalizability of context-aware delimiter classification to unseen legal corpora — a relevant consideration for the present work, where the Brazilian Portuguese legal register introduces additional boundary ambiguity through citation strings, procedural enumerations, and multi-clause sentence structures. Almalki et al. [03_03] propose a segmentation-and-aggregation framework for legal judgment prediction in Arabic, in which documents are segmented into sentence-level inputs, independently classified with AraBERT, and then aggregated to produce document-level decisions. Their success in overcoming the 512-token transformer limit through sentence-level decomposition provides a structural precedent for the LLM-assisted segmentation approach adopted in Stage3 of the present pipeline.

Ragazzi et al. [03_04] introduce LAWSUIT, a large-scale dataset for abstractive summarization of Italian constitutional court verdicts, and demonstrate that segmentation-based summarization pipelines outperform end-to-end sequence-to-sequence models across both full and few-shot settings. Their finding — that the uniform distribution of salient content throughout Italian legal verdicts prevents position-based shortcuts — resonates with the challenge encountered in Brazilian criminal labor-law decisions, where legally critical information (the conviction, the penalty, and the parties' identities) is similarly distributed across structurally heterogeneous document sections. Chen et al. [03_06] independently confirm the superiority of hierarchical segmentation in the context of long-document summarization using LLMs, showing that combining document decomposition with Chain-of-Thought (CoT) prompting significantly reduces factual hallucination and improves coverage across scientific, governmental, and legal domains. Their ablation studies confirm that segmentation and reasoning guidance make independent, complementary contributions to output quality.

2.3. Named entity recognition in legal and low-resource settings

Legal NER is the most direct antecedent to Stage4 of the present pipeline. Wang et al. [03_02] propose AT-Flat, an adversarial training model that integrates task-shared word boundary signals from Chinese Word Segmentation (CWS) into a nested NER architecture for Chinese judicial documents. By combining character-word fusion via the Flat-Lattice Transformer with an adversarial multi-task framework, they address the nested entity structure that is also present in Brazilian legal text, where party names, judicial organizations, and statutory references frequently overlap. Their demonstration that adversarial boundary transfer improves NER performance in specialized legal corpora — evaluated on traffic accident and financial lending judgment datasets — validates the importance of domain-aware architectural design for legal NER.

Bellandi et al. [01_01] present an entity-centric infrastructure for Italian court judgments that combines machine learning and syntactic rule-based NLP services in an iterative annotation pipeline with a human-in-the-loop validation layer. Their co-design methodology, involving legal institutions and implementation partners alongside a scientific team, and their emphasis on building a sustainable, multi-purpose annotation infrastructure rather than a single-task

system, closely mirrors the design philosophy of the present framework. The scalability challenges they identify — particularly the difficulty of maintaining annotation quality at national scale with limited expert involvement — are directly relevant to the Brazilian context, where the volume of labor-court decisions is substantial and expert annotation resources are scarce. Sakhaee and Wilson [01_02] demonstrate that information extraction approaches based on rule-based identification of citation expressions can achieve precision and recall exceeding 98% when constructing legislation networks from large, noisy legal corpora. Their success in handling poor-quality scanned documents through approximate string matching underscores the practical value of preprocessing robustness — a principle instantiated in Stage1 of the present pipeline through aggressive PJe footer removal and document-type normalization.

Wang et al. [03_09] propose CLWKD, a cross-lingual semantic alignment framework that leverages multi-granularity mapping matrices and knowledge distillation to transfer domain-specific representations from high-resource languages to low-resource agglutinative languages. While developed for Mongolian-Chinese and Korean-Chinese NLP tasks, the framework's motivation — overcoming the joint challenges of domain-specific data scarcity and morphological complexity — is structurally analogous to the resource constraints faced in Brazilian Portuguese legal NLP, where both domain-adapted models and annotated legal corpora remain limited relative to English counterparts. The paper establishes cross-lingual knowledge distillation as a promising paradigm for extending legal NLP capabilities to underrepresented languages without requiring large-scale domain-specific annotated corpora.

Zhu et al. [01_03] design a knowledge graph retrieval system for Chinese public prosecutorial affairs, combining a bi-LSTM-CRF NER module with ontology-based graph storage and latent semantic indexing. Their multichannel input strategy — fusing text representations, word vectors, and entity vectors — and their integration of NER within a broader knowledge management infrastructure provide a relevant reference for the entity aggregation and output schema design in Stage4 of the present work. Similarly, Guha et al. [03_08] demonstrate that multi-modal document classification — combining VGG16 image features with Legal-BERTbase text representations — achieves F1-scores exceeding 97% for document boundary detection in a Title Insurance context. Their confirmation that textual features are the primary determinant of classification performance, with visual features providing marginal but operationally significant gains, reinforces the text-centric design of the present pipeline.

2.4. Comparative synthesis

Table 1 summarizes the most closely related studies, positioning each work with respect to the dataset, language, primary task, model family, and reported metrics. The comparison highlights the principal research gap addressed by the present study: no prior work has proposed a multi-layer, end-to-end legal NLP framework for Portuguese-language criminal labor-law decisions that jointly addresses document-type filtering, penal-merit classification, structured segmentation, and named-entity extraction within a single, cohesive pipeline. Existing pipelines are either single-stage, language-specific (predominantly English or Chinese), or focused on a subset of the tasks addressed here.

3. Dataset and Problem Definition

3.1. Corpus description

- Raw corpus size: 2,581 documents.
- Document heterogeneity (sentença, acórdão, despacho, ata, etc.).

3.2. Task definitions

- T1: Document-type filtering.
- T2: Penal-merit classification (classes: condenação, absolvição, extinto).
- T3: Decision segmentation.
- T4: Named-entity extraction.

3.3. Annotation and label governance

- Define legal criteria for label assignment (Art. 149 framing).
- Report annotation process and expert validation protocol.

3.4. Data governance and ethics

- LGPD/privacy handling, anonymization strategy, and legal-compliance statement.

4. Methodology

4.1. Overview of the multi-layer pipeline

Figure 1 presents the end-to-end workflow adopted in this study. The process starts with document-type normalization and filtering (Stage1_1), keeping only judicial sentences that are relevant to criminal-merit analysis. Next, Stage1_2 applies task-adaptive preprocessing and generates two normalized outputs: one optimized for merit classification and one that preserves legal-text patterns for segmentation and NER. Stage2 performs penal-merit classification, Stage3 applies LLM-assisted segmentation to the conviction subset, and Stage4 performs legal-domain named-entity extraction over segmented content.

4.2. Stage1_1: Document-type normalization and filtering

- Pattern-based extraction of document type from metadata-like spans.
- Final retention of *sentença* documents.

Table 1
Comparative summary of related works. SBD = Sentence Boundary Detection; RRL = Rhetorical Role Labeling; LJP = Legal Judgment Prediction; NER = Named Entity Recognition; AS = Abstractive Summarization; DSS = Document Stream Segmentation; F1 = macro/weighted F1-score; Acc = accuracy.

Study	Ref.	Lang.	Task	Model family	Metric	Key limitation
Bellandi et al. (2024)	[01_01]	IT	NER / Annotation framework	ML + rule-based NLP	Feasibility	Single language; limited labelled data
Sakhaee & Wilson (2021)	[01_02]	EN	Citation extraction / legislation network	Rule-based IE	98% P&R	No learning; specific to NZ legislation
Zhu et al. (2022)	[01_03]	ZH	NER / Knowledge graph	BiLSTM-CRF + Neo4j	Acc (crime pred.)	Single domain; Chinese only
Resck et al. (2023)	[02_01]	PT	Citation classification	ML + interpretability	96% Acc	Single court; binary task
Ariai et al. (2025)	[02_02]	Multi	Survey (LTC, NER, LJP, etc.)	Survey	—	Survey; no new experiments
Nuranti et al. (2025)	[02_03]	ID	NER corpus construction	BiLSTM-CRF / BIO annotation	IAA 0.705	Method paper; no trained model reported
Wang et al. (2025)	[03_02]	ZH	Nested NER in legal text	AT-Flat (FLAT + adversarial)	State-of-the-art F1	Chinese only; long-doc limitations
Almalki et al. (2025)	[03_03]	AR	LJP / segmentation-aggregation	AraBERT + aggregation	85.62% Acc	Binary outcome; Arabic only
Ragazzi et al. (2025)	[03_04]	IT	Legal abstractive summarization	Seq2seq + segmentation	ROUGE	Summarization only; Italian only
Bhattacharya et al. (2023)	[03_05]	EN	Rhetorical role labeling	Hier-BiLSTM / BERT / Legal-BERT	F1 0.82 (IN)	EN/IN focus; no PT variant
Chen et al. (2025)	[03_06]	Multi	Long-doc summarization	CoT + hierarchical LLM	ROUGE / FactCC	High compute; summarization only
Sheik et al. (2025)	[03_07]	EN	Sentence boundary detection	CNN + CRF hybrid	F1 +4%	English legal text; no multilingual eval
Guha et al. (2022)	[03_08]	EN	Document stream segmentation	VGG16 + Legal-BERT (multimodal)	F1 97.37%	Proprietary data; TI domain only
Wang et al. (2026)	[03_09]	ZH/MN/KO	Cross-lingual domain alignment	CLWKD (mapping + KD)	Cross-lingual NLP	Low-resource focus; no legal-specific eval
This work	—	**PT**	**Filtering + classification + segmentation + NER**	**Legal-BERT + classical + LLM + NER**	**F1 / pipeline**	**Low-resource; single court system**

4.3. Stage1_2: Task-adaptive legal text preprocessing

- PJe duplicate-footer removal.
- Aggressive normalization for classification output.
- Conservative normalization for NER/segmentation output.

4.4. Stage2: Penal-merit classification with Legal-BERT embeddings

- Encoders avaliados: `dominguesm/legal-bert-base-cased-ptbr`, `neuralmind/bert-base-portuguese-cased` e `neuralmind/bert-large-portuguese-cased`.
- Classifiers: Logistic Regression, linear SVM, Random Forest, XGBoost.
- Stratified k-fold protocol and weighted metrics.

4.5. Stage3: LLM-assisted segmentation of conviction decisions

- Input restricted to conviction subset.
- Prompting protocol and JSON schema validation.
- Retry/failure handling and quality-control checks.

4.6. Stage4: Legal-domain NER extraction

- Model: `dominguesm/legal-bert-ner-base-cased-ptbr`.

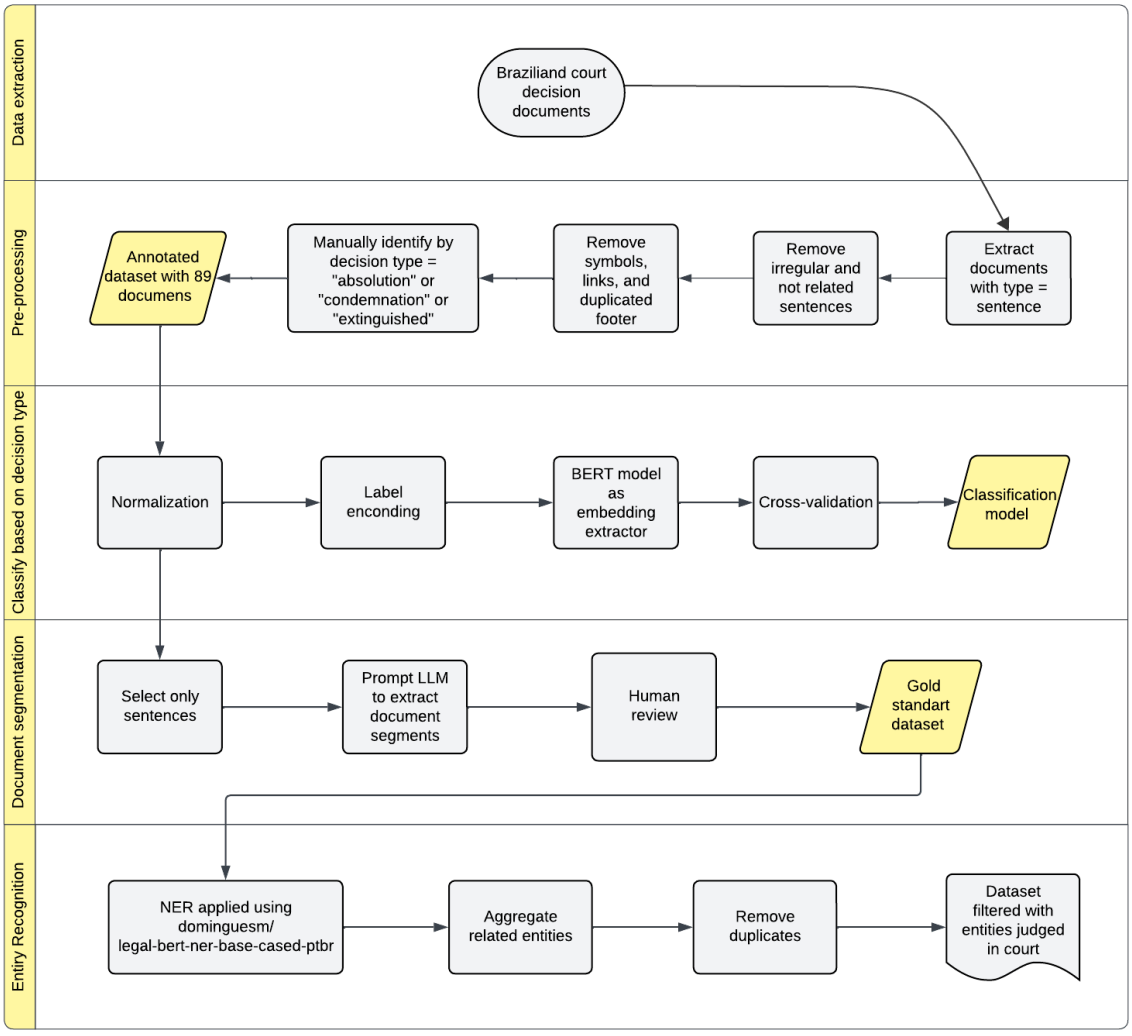

Figure 1: Activity diagram of the proposed multi-layer pipeline for legal document filtering, penal-merit classification, segmentation, and NER extraction.

- Entity aggregation strategy and output schema.

5. Experimental Setup

5.1. Evaluation protocol

- Data splits, fold strategy, seeds, and confidence intervals.
- Primary metrics: macro-F1, weighted-F1, precision, recall, per-class scores.

5.2. Baselines and ablations

- Baseline without multi-layer filtering.
- Baseline with single normalization stream.
- Ablation removing each stage (one-at-a-time).

5.3. Implementation details

- Software stack, hardware profile, runtime, and cost profile (LLM calls).

6. Results

6.1. Merit classification performance

Table 2 compares three encoder backbones under the same downstream classifiers, using weighted F1 as the primary criterion. The best overall result is highlighted in bold and corresponds to BERTimbau-large with linear SVM ($F1 = 0.6152 \pm 0.0808$). Legal-BERT remains competitive and obtains the strongest XGBoost result ($F1 = 0.6114 \pm 0.0871$), while BERTimbau-base shows gains for Logistic Regression and Random Forest.

In practical terms, this indicates that domain adaptation in the encoder remains beneficial for the XGBoost configuration, while larger general-purpose Portuguese encoders can provide stronger performance with linear classifiers. Given this trade-off, both Legal-BERT+XGBoost and BERTimbau-large+SVM should be considered viable Stage2 operating points depending on interpretability and computational constraints.

Table 2
Stage2 encoder comparison across downstream classifiers (weighted F1, mean ± std across folds).

Classifier	Legal-BERT (dominguesm)	BERTimbau-base (neuralmind)	BERTimbau-large (neuralmind)
XGBoost	0.6114 ± 0.0871	0.5261 ± 0.0590	0.5661 ± 0.0558
SVM (Linear)	0.5415 ± 0.0484	0.5350 ± 0.0664	$\mathbf{0.6152 \pm 0.0808}$
Random Forest	0.5406 ± 0.1249	0.5611 ± 0.0622	0.5284 ± 0.0420
Logistic Regression	0.5399 ± 0.0522	0.5627 ± 0.0613	0.6136 ± 0.0799

TODO: Mention that the BERT models were not fine-tuned for this specific case due to the limited amount of available data.

6.2. Segmentation quality

Report schema-valid rate, section-level agreement, and representative errors.

6.3. NER extraction quality

Report entity-level precision/recall/F1 (strict and relaxed matching if possible).

6.4. End-to-end pipeline gains

Compare extraction quality before vs. after multi-layer filtering.

7. Discussion

7.1. Why the pipeline works
7.2. Failure modes and error taxonomy
7.3. Generalization to other legal domains/languages
7.4. Operational and policy implications

8. Threats to Validity

- Internal validity: label noise, annotation drift, prompt sensitivity.
- External validity: transfer to other courts, years, and jurisdictions.
- Construct validity: legal meaning vs. NLP proxy metrics.

9. Reproducibility Statement

- Public code structure, dependency pinning, random seeds, and artifact inventory.
- Which resources can/cannot be publicly shared due to legal constraints.

10. Conclusion and Future Work

- Summarize empirical and practical contributions.
- Future work: active learning, expert-in-the-loop verification, and cross-court transfer evaluation.

Acknowledgments

Funding and institutional acknowledgments.

CRediT authorship contribution statement

Carlos A. V. Tinin: Conceptualization, Methodology, Software, Writing – original draft. **Edison P. de Freitas:** Validation, Formal analysis, Writing – review & editing. **Kevin H. M. Gularte:** Validation, Formal analysis, Writing – review & editing. **Elena J. da Costa:** Validation, Formal analysis, Writing – review & editing. **João Paulo J. da Costa:** Validation, Formal analysis, Writing – review & editing.

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